





Recap

▪ Generated Value

- firms generate value when *value* of produced goods exceeds *cost* of resources used up
- microeconomics: in well-functioning markets economic agents make free choices:
 - *selling prices* – lower bound of *utility* they give to consumers who purchase them
 - *purchase prices* – upper bound of *cost* incurred by producers who manufacture them or subjective value of employees providing *labor*

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Recap

▪ Accrual Income

- accounting reflect microeconomic substance if:
 - *revenues* reflect value truly created
 - *expenses* capture all economic costs incurred
- accounting principles (rules)
 - *revenues* – when and how much value can a firm claim to have created in fiscal period?
 - *expenses* – what are all corresponding costs used up to generate the value?

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Recap

- **Revenue Recognition**

- two parties involved in a deal:
 - *earned (seller)* – firm must deliver goods or render services (completed)
 - *realized or realizable (buyer)* – customer must pay in cash or provide assets *or* promise to do so in the future (completed or expected)
- note that condition is:
 - *stronger* for the firm (the seller)
 - *weaker* for the customer (the buyer)

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Recap

- **Timing**

- typically, revenues are *realizable* already at the time of signing the legal contract
- thus, revenues are typically recognized when *earned*, i.e., at the time of delivery

- **Exceptions**

- *before* – percentage of completion, in *large* project for informative depiction of reality
- *after* – when high *uncertainty* about warranty costs, rights of return, or customer payments

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Recap

- **Expense Matching**
 - *revenues* must be *matched* with all corresponding *expenses* incurred in generating revenues
 - *matching* of revenues expenses implies that periodic earnings show *value created*
 - *mis-match* between positive and negative flows is tolerated in SCF but not in IS
- **Expense Classification**
 - different expenses vary in *relevance* for evaluating firm performance
 - decomposition of expenses matters (K&L)

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Recap

- **Violation of the Matching Principle**
 - *uncertain future benefit*
 - *expense* today ~ future *revenue*
 - cash paid for R&D, advertising, restructuring is today *expensed* as incurred rather than *capitalized* as assets and amortized
 - *insufficient past expense*
 - *expense* today ~ past *revenue*
 - assets depreciated too slowly are *written off* today, insufficient provisions for lawsuit charges imply today's expenses

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Inventory



Balance Sheet	
cash	notes payable
marketable securities	current maturities of l-t debt
accounts receivable	accounts payable
inventories	accrued liabilities
prepaid expenses	unearned revenue
	long-term debt
	deferred taxes
property, plant & equipment	contributed capital
intangible assets	retained earnings

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Inventory



Inventory

“Inventory consists of the goods that a company legally owns and expects to sell for a profit in the course of normal operations. In other words, it’s the products the company owns and intends to sell.”

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Inventory

Inventory

“A current asset whose ending balance should report the cost of a merchandiser’s products [..]. The inventory of a manufacturer should report the cost of its raw materials, work-in-process, and finished goods. [It] should include all costs necessary to acquire the items and to get them ready for sale.”

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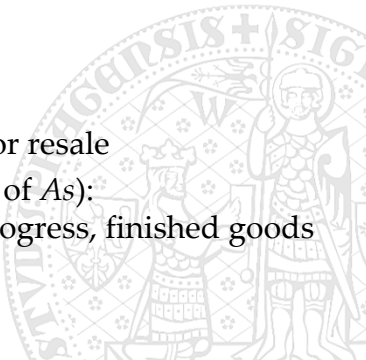
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Inventory



- **Definition**
 - items held for sale or used in manufacturing of products that will be sold in the course of normal business
- **Components**
 - *retail* (45% – 65% of As): merchandise purchased for resale
 - *manufacturing* (25% – 35% of As): raw materials, work-in-progress, finished goods
 - *services*: ---



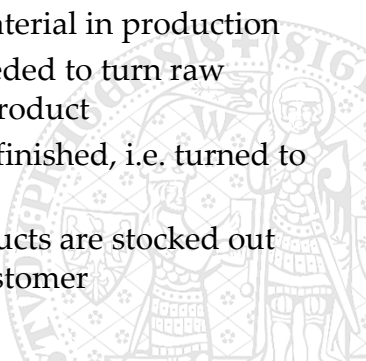
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Inventories



- **Accounting for Inventory**
 - record following inventory transactions:
 - buy raw materials for **\$300**
 - start using **1/3** of the material in production
 - labor cost of **\$50** are needed to turn raw material into finished product
 - **all** work-in-progress is finished, i.e. turned to finished products
 - **1/2** of the finished products are stocked out and delivered to the customer



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Inventories

▪ Accounting for Inventory

Balance Sheet				Income Statement	
Assets		Liabilities, Equity		Expenses	Revenues
cash	1 300				
raw material	1 300 2 100				
work in progress	2 100 3 50 4 150	salaries payable	3 50		
finished goods	4 150 5 75			COGS	
				5 75	

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Inventory

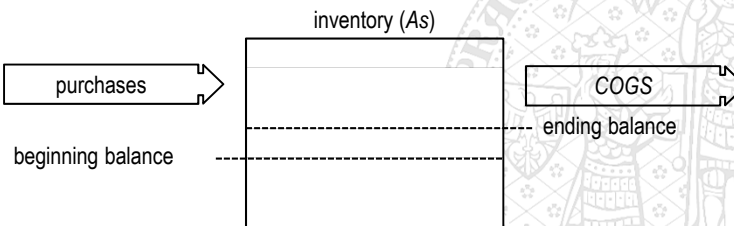
Income Statement	
+	net sales
-	cost of goods sold (COGS)
	gross margin
-	selling, general and administrative expenses (SG&A)
+/-	other operating income/expense
	operating income (EBIT)
-	interest expense
+/-	other financial income/expense
	pre-tax income
-	income tax expense
	net income (NI)

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Cost Allocation

- **Distribution of Units**
 - inventories are purchased at various (often rising) input prices
 - we must decide which units are used in production and which are still kept in warehouse



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Cost Allocation

- **Cost Distribution**
 - costs of goods available for sale must be distributed between:
 - ending inventory (Iv_t), which stays on BS
 - cost of goods sold ($COGS_t$), which affects NI_t



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Cost Allocation

- **Specific Identification Method**
 - very precise but rather costly to implement
 - used only rarely (e.g. antiques, second-hand cars)
- **Cost Flow Assumption**
 - keep track of individual unit's costs would be cumbersome or even impossible (e.g. crude oil)
 - assumption is made about the order in which purchased unit are used in production
 - the assumption may not correspond with actual physical stocking out of individual units

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Cost Flow Assumption

- **FIFO**
 - first-in-first-out
 - the oldest units are used first
- **LIFO**
 - last-in-first-out
 - the newest units are used first
- **Average Cost**
 - average cost of all units currently in warehouse

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Cost Flow Assumption

▪ Example

- a company purchased and used its inventory in production in the following sequence:
 - buy 10 units at **\$10/unit**
 - buy 10 units at **\$12/unit**
 - buy 10 units at **\$15/unit**
 - use 10 units in production
 - use 5 units in production
 - buy 10 units at **\$20/unit**
 - use 10 units in production

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Cost Flow Assumption

FIFO
“pipeline”

\$20 / unit
\$15 / unit
\$12 / unit
\$10 / unit

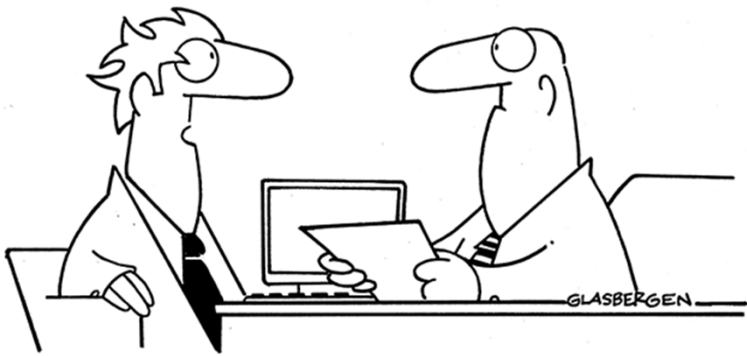
LIFO
“reservoir”

\$20 / unit
\$12 / unit
\$10 / unit

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Inventory



GLASBERGEN

"I don't have any experience counting inventory, but I'm pretty good at guessing how many jellybeans are in a jar."

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Cost Flow Assumption

- **Questions**
 - What are *pros & cons* of LIFO & FIFO?
 - Which is more representationally *faithful*?
 - Where is the *bias* more problematic?
 - If given a free choice what would firms choose
 - in *financial* accounting
 - in *tax* accounting
 - Is that choice allowed by accounting standards?
 - Why does IFRS forbid LIFO?
 - How can a firm manipulate accounting in LIFO?
 - How to mitigate possible investors' confusion?

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Cost Flow Assumption

- **Measurement Bias**
 - when prices rise (inflation) *recently* bought goods best reflect their current replacement cost
 - FIFO ~ “LILO” – recent units in *Iv* (BS):
 - *benchmark* inventory measurement
 - LIFO – recent units in *COGS* (IS):
 - *accurate* gross profit – fair performance measure
 - *understated* inventory – “hidden reserve”

	gross profit (IS)	inventory (BS)
FIFO	overstated	OK
LIFO	OK	understated



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Cost Flow Assumption

- **Managerial Incentives**
 - managers would prefer:
 - FIFO in *financial accounting* to report higher gross profit $\subset$ net income (“dating game”)
 - LIFO in *tax accounting* to lower taxable income (“begging game”)
 - such a combination would be opportunistic and so *restrictions* on the method choice are needed

	gross profit (IS)	inventory (BS)
FIFO	overstated	OK
LIFO	OK	understated

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Cost Flow Assumption

- **IFRS**
 - LIFO *forbidden*
 - non-U.S. firms choose between FIFO and AC
- **U.S. GAAP**
 - LIFO *allowed* but conformity between tax and financial accounting required
 - most companies use LIFO to lower taxes, they *must* use LIFO also in financial accounting



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Cost Flow Assumption

- **Why LIFO?**
 - provides accurate measurement of *COGS* in IS, which is arguably the more important financial statement relative to BS
- **Why FIFO?**
 - typically reflects *physical flow* of inventory (e.g. fresh milk), but for raw materials (e.g. coal)
 - *temporary manipulation* – persistent bias in *COGS* may be less bad than one-off manipulation through liquidation of LIFO reserve

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Cost Flow Assumption

▪ LIFO Shortcomings

- *physical movement* – firms tend to use oldest raw material first (~ FIFO)
- *accounting manipulation* – understated inventory value on BS implies a “LIFO reserve” that can be opportunistically used to temporarily improve NI

\$22 / unit	
\$12 / unit	} cheap “old” inventory
\$10 / unit	

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Cost Flow Assumption

▪ Limit Impact?

- How to limit confusion from temporary income manipulation by liquidation of LIFO reserve?

▪ Compute Effect?

- How to compute magnitude of income improvement by liquidation of LIFO reserve?

▪ Trust Investors?

- Decision on LIFO depends on the trust in investors’ sophistication and meticulousness.

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Cost Flow Assumption

- **Disclosure**
 - *inventory method* – must be disclosed in footnotes, e.g.
 - “Inventories are carried at the lower of the cost (LIFO) or market.”
 - *LIFO reserve* – difference in inventory value under current method and benchmark method (i.e. FIFO) must also be disclosed, e.g.
 - “If the FIFO method of inventory accounting had been used, inventories would have been approximately \$... higher than reported.”

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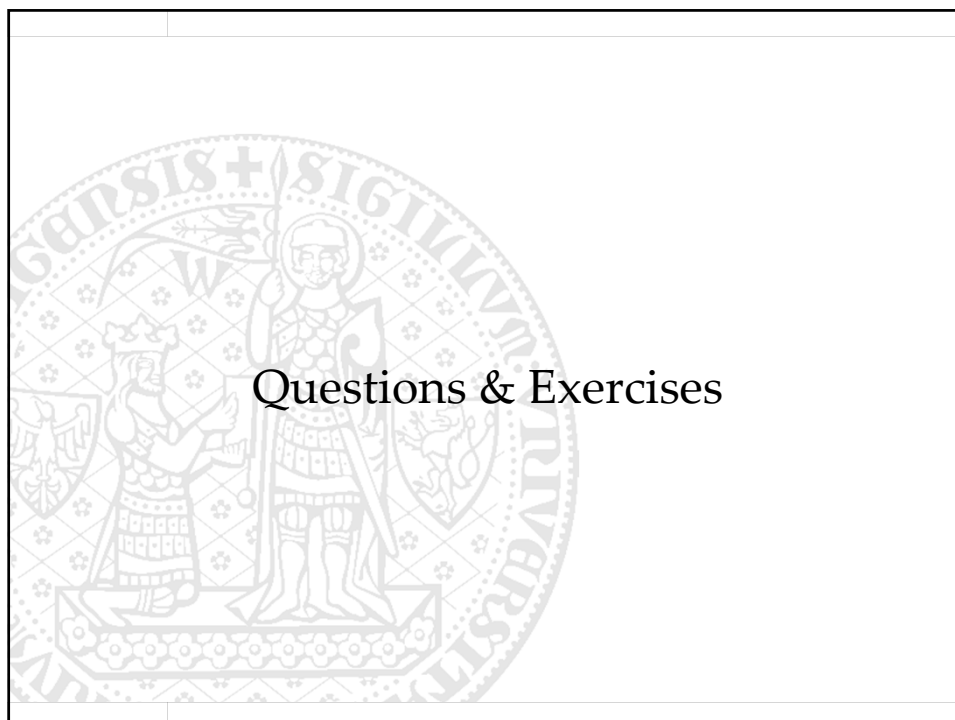
Difficult Profitability ...



“We’re looking for people who can help make this company profitable again. I’ll read your resumé for \$500.”

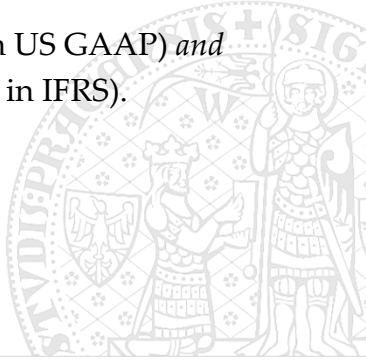


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Questions

- **FIFO vs. LIFO**
 - FASB and IASB have clearly different views on pros and cons using FIFO and LIFO.
 - Discuss arguments:
 - for allowing LIFO (as in US GAAP) *and*
 - for forbidding LIFO (as in IFRS).



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Exercise 3a

- **Net Purchases of BLSH**

- A merchandizing firm “Buy Low, Sell High Ltd.” (BLSH) is a retail company selling clothes. The value of its inventory was \$ 1 100 at the end of 2000 and \$ 1 300 at the end of 2001. During 2001 BLSH generated net sales of \$ 3 600, which matching COGS of \$ 3 100 including \$ 600 in salaries (ignore all other costs).
- What were the net purchases of inventory in 2001?

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Exercise 3b

- **Cost Flow Assumption**

- a company purchased and used its inventory in production in the following sequence:
 - buy 10 units at \$10/unit
 - buy 10 units at \$12/unit
 - buy 10 units at \$15/unit
 - use 10 units in production
 - use 5 units in production
 - buy 10 units at \$20/unit
 - use 10 units in production

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Exercise 3b

- **Required**

- What is the value of remaining inventory under FIFO, LIFO, and AC?
- What were the COGS of the period under FIFO, LIFO, and AC?
- What is the total cost of purchased inventory under FIFO, LIFO, and AC?
- As a manager which method would you choose to (1) best reflect the physical flow of goods, (2) to minimize taxes?

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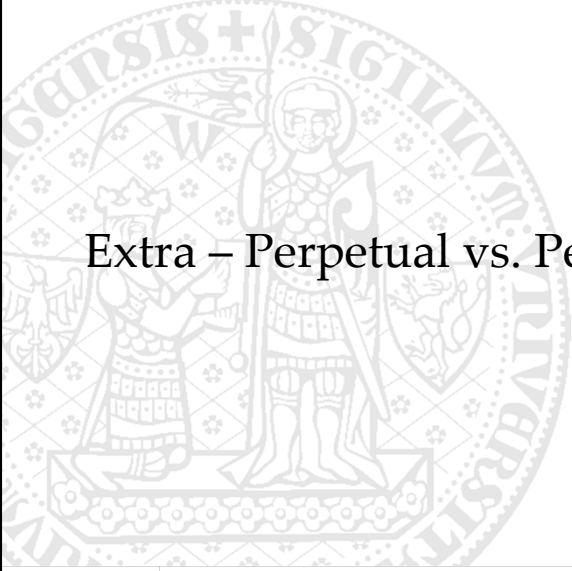


Exercise 3b

- **Required**

- What is the “LIFO reserve” under FIFO, LIFO, and AC?
- Assume the firm uses LIFO and delays the purchase of the last batch (at \$20/unit) till next period. How will this affect its LIFO reserve, COGS and NI?

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Extra – Perpetual vs. Periodic System



Inventory Systems

- **Perpetual System**
 - keeps a running, continuous record of inventories and COGS that is updated after every transaction
 - purchases are recorded as increases of inventory, COGS are recognized at the time of sale
 - adjustment at the end of the accounting period if the physical count differs from the ending balance of the inventory account (?)
 - useful to control flow of materials or merchandise and to aid in controlling inventory losses



Inventory Systems

- **Periodic System**

- computes the COGS periodically based on physical counts rather than after each sale
- acquisitions are recorded at a temporary account called “purchases”; ending balance of inventory is determined by physical count; COGS are determined based on:

$$COGS_t = Iv_{t-1} + purchases_t - Iv_t$$

- cheaper to implement; allows to maximize the tax benefits due to LIFO valuation method
- no updated record of units sold or on hand

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Extra - Sales Returns & Allowances



Sales Returns

- **Occurrence**

- customers can cancel their purchase or request a price concession in case the delivered goods are damaged or do not meet the pre-contracted parameters
- this is essentially a reduction in previously contacted sales and that is also how it is treated

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Sales Returns

- **Treatment**

- “sales returns and allowances” (*SRA*) is a contra-revenue account offsetting sales (that is why we refer to “net sales”)
- the treatment to bad debts, but for the fact that the returned goods increase inventory and thereby decrease *COGS*
 - *AR* or cash are credited (reduction in cash or receivables), *SRA* is debited (reduction in sales)
 - *Iv* are debited (goods returned), *COGS* are credited (reduction in matching costs)

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Sales Returns

Transaction Analysis

- ① merchandise purchased for \$ 720 sold to a customer for \$ 800 (no sales returns expected)
- ② one quarter of the goods is returned because it has been shipped to a wrong customer

Assets		Liabilities, Equity	Expenses		Revenues		
Accounts Receivable			COGS		Sales		
❶	800		❶	720		❶	800
	❷		200		❷	180	
Inventories					SRA		
	❶	720			❷	200	
❷	180						

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Sales Returns

Allowance for Sales Returns and Allowances

- ① at the time of sale estimated *SRA* is recorded and an contra-asset allowance for sales returns and allowances (*ASRA*) is created
- ② when goods are returned *AR* are eliminated against *ASRA* (i.e. no IS effect)

Assets		Liabilities, Equity	Expenses	Revenues	
	Accounts Receivable			Sales	
①	800				① 800
②		30			
	ASRA			SRA	
①		40		①	40
②	30				
② ①	760				

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